



國立陽明交通大學資訊工程學系資訊中心

IT Center of Department of Computer Science, National Yang Ming Chiao Tung University

System Maintenance and SFTP

Computer System Administration Homework 2

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Outline

- preflight: initramfs, LUKS and systemd.target
- 2-1: LVM resize, BTRFS on LVM & BTRFS snapshot
- 2-2: Manage SFTP file server
- 2-3: SFTP logging and Monitoring daemon

Preflight

- Create a VM base on the provided [vmdk file](#)
 - Pick the one that works on your machine
 - Mount the vmdk as **/dev/sda**
- You may encounter some issues when boot up the image we provided
 - The initramfs image is under **/root**. Try to find and use it to fix your machine.
 - After booting up, you will land in rescue mode. Try to find your way out.
 - **DO NOT** reinstall the VM or use mkfs against any partition.

Preflight

```
Booting `Debian GNU/Linux'

Loading Linux 6.12.107+deb13-arm64 ...
Loading initial ramdisk ...
error: file `/initrd.img-6.12.107+deb13-arm64' not found.

Press any key to continue...
[ 3.834498] /dev/root: Can't open blockdev
[ 3.838751] List of all bdev filesystems:
[ 3.839485] fuseblk
[ 3.839536]
[ 3.840336] Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0)
[ 3.841984] CPU: 0 UID: 0 PID: 1 Comm: swapper/0 Not tainted 6.12.107+deb13-arm64 #1 Debian 6.12.107-1
[ 3.843257] Hardware name: QEMU QEMU Virtual Machine, BIOS 4.2026.08-1 08/26/2026
[ 3.844523] Call trace:
[ 3.844878] dump_backtrace+0xd8/0x130
[ 3.845893] show_stack+0x20/0x38
[ 3.846257] dump_stack_lvl+0x60/0x80
[ 3.846890] dump_stack+0x18/0x28
[ 3.847406] panic+0x164/0x378
[ 3.847701] mount_root_generic+0x200/0x2c8
[ 3.848072] mount_root+0x17c/0x1a8
[ 3.848443] prepare_namespace+0x1e0/0x250
[ 3.848809] kernel_init_freeable+0x2c4/0x2f0
[ 3.849159] kernel_init+0x28/0x150
[ 3.849467] ret_from_fork+0x10/0x20
[ 3.850404] SMP: stopping secondary CPUs
[ 3.852001] Kernel Offset: disabled
[ 3.852464] CPU features: 0x08,80002023,c0200000,0200421b
[ 3.853041] Memory Limit: none
[ 3.854110] ---[ end Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0) ]---
```

Preflight

- The default disk encryption password is `judge`
 - You should change it to `ilovesa2026`
- Make sure the VM can pass HW0
 - No passwd sudoers
 - wireguard (***Remember to use the `second ip` of the tool***)
 - ssh key
 - hostname
 - (you do not have to resubmit it, just make sure the config stays the same)
- Boot with UEFI mode

Preflight:Hint

- Choose Debian when prompted to select the OS type.
- Import [vmdk](#) file
 - Download the vmdk file corresponding to the instruction set your cpu support (i.e. aarch64/amd64)
 - You may need to convert the image into different format (e.g. qcow2)
 - ***If your CPU support neither x86_64 nor arm64, please contact TA ASAP***

Preflight:Hint

- default user and password: judge/judge
- root passwd: judge
- [visudo\(8\)](#), [sudoers\(5\)](#), NOPASSWD
- LUKS, [dm-crypt](#)
- [ramfs, rootfs and initramfs](#)
- [systemd.target\(5\)](#)
- [find\(1\)](#)

(2-1) (48%)

LVM resize, BTRFS on LVM
And BTRFS snapshot

HW2-1: Overview

- Shrink root LV partition
- Setup **LVM**, and setup **RAID10** with LVM
- Make **BTRFS** on RAID10 LVM
- Manage BTRFS with **flatten layout**
- Write a **SnApper** script to **manage BTRFS snapshot**
 - create, list, delete, rollback

HW2-1: Requirements (1/12)

- Shrink debian-vg-root to 15G
- Create a new LV with the rest of VG and
 - named the LV as **new-part**
 - Format the LV with ext4 filesystem
 - Mount the LV to `/mnt/new-part` (should persist after reboot)

HW2-1: Requirements (2/12)

- Add at least one disk
- **Make 4 PV** from added disks and **group** these 4 PV into a VG
 - named the VG as **SA2026vg**
- **Create a LV** on top of VG
 - named the LV as **SARaid10**
 - Make the LV as **Raid10**
- **Make BTRFS filesystem** on `/dev/mapper/SA2026vg-SARaid10`
- Set the **Label** of the BTRFS filesystem to **SAHW2**

HW2-1: Requirements (3/12)

- All of your **subvolume** should **directly under root volume**
 - root volume: subvolume id = 5
 - **No nested subvolume is allowed** (subvolume under other subvolume)
- Create following **subvolumes** under root volume:
 - **root, sftp, pool1, pool2**
- Create following **directories** under **root volume**:
 - **snapshot/sftp, snapshot/pool1, snapshot/pool2**

NOTE1: **root** is a **subvolume named root**. While **root volume** is volume **with id 5**. They are different.

HW2-1: Requirements (4/12)

- Mount subvolumes according to the table
 - All the mounting **source device** should be `/dev/mapper/SA2026vg-SARaid10`
 - All the mounting should be **persistent**
 - i.e. in `/etc/fstab`

Subvolume name	Mount target
root	<code>/mnt/hw2</code>
sftp	<code>/mnt/hw2/sftp</code>
pool1	<code>/mnt/hw2/pool1</code>
pool2	<code>/mnt/hw2/pool2</code>

NOTE1: **root** is a subvolume named **root**. While **root volume** is volume with id 5. They are different.

HW2-1: Requirements (5/12)

- Write a BTRFS snapshot management script. Named as **SnApper**
- Implement following function:
 - **Create**: creating a snapshot for a subvolume
 - **List**: listing all snapshot managed by this script
 - **Delete**: remove a snapshot from the system
 - **Rollback**: Apply a snapshot onto subvolume
- The **SnApper** should resides in your system `$PATH`
- You can use anything to implement **SnApper**
 - But we suggest using bash script

HW2-1: Requirements (6/12)

- SnApper should show following help message

```
Usage:  
SnApper [-h]: show this message  
SnApper snapshot SUBVOL [-c ROTATION_COUNT]  
SnApper list [-p SUBVOL] [-i ID]  
SnApper delete [ID]  
SnApper rollback ID
```

HW2-1: Requirements (7/12)

- **SnApper snapshot SUBVOL [-c ROTATION_COUNT]**
 - create a snapshot for **SUBVOL**
 - The created snapshot should also be a subvolume **directly under root volume**
 - The snapshot should be placed in directory **<ROOT_VOLUME>/snapshot/<SUBVOL>** and named as **@<YYYYMMDD-hhmmss>**

```
$ sudo SnApper snapshot pool1
Snap 'snapshot/pool1/@20251003-163650' [551]
$ sudo btrfs subvolume list /mnt/hw2/ -p -a -t
```

ID	gen	parent	top level	path
--	---	-----	-----	----
...				
551	671	5	5	<FS_TREE>/snapshot/pool1/@20251003-163650

HW2-1: Requirements (8/12)

- After create, make sure snapshot of *SUBVOL* doesn't **exceed** *ROTATION_COUNT*
 - if *-c* is not provided, the rotation count is **default to 5**
 - if exceed, **delete** snapshot with **smallest subvolume ID**
 - repeat until snapshot count $\leq$ *ROTATION_COUNT*
- The command should **output on success**:
 - `Snap '<SNAPSHOT_PATH>' [<ID>]`
- Created snapshot should be **readonly**

HW2-1: Requirements (9/12)

- **SnApper list [-p SUBVOL] [-i ID]**
 - If no **-p** and **-i** provided, **listing all snapshot** managed by **SnApper**
 - If **-p** is provided, only show snapshot of **SUBVOL**
 - If **-i** is provided, only show snapshot with that **ID**
 - **-p** and **-i** can be both provided
 - Do not use TAB to separate the column

```
$ sudo SnApper list
ID    SUBVOLUME  TIME
549    pool1      2025-10-02 20:57:50
553    pool2      2025-10-03 17:23:17
$ sudo SnApper list -i 549
ID    SUBVOLUME  TIME
549    pool1      2025-10-02 20:57:50
$ sudo SnApper list -i 549 -p pool2
ID    SUBVOLUME  TIME
```

HW2-1: Requirements (10/12)

- **SnApper delete [ID]**
 - delete the snapshot with provided **ID**
 - if **ID** is not provided, **delete all snapshot**
- The command should output on success:
 - **Destroy ID <ID>**

```
$ sudo SnApper delete 549
Destroy ID 549
$ sudo SnApper delete
Destroy ID 554
Destroy ID 555
Destroy ID 556
```

HW2-1: Requirements (11/12)

- **SnApper rollback *ID***
 - Rollback to snapshot specified by *ID*
 - *ID* must be provided
 - You should automatically find corresponding subvolume of the snapshot
- The command should output on success
 - Rollback '<SNAPSHOT_PATH>' [<ID>] to <SUBVOL>

```
$ sudo SnApper list
ID    SUBVOLUME  TIME
557   pool2     2025-10-03 19:45:29
$ sudo SnApper rollback 557
Rollback 'snapshot/pool2/@20251003-194529' [557] to pool2
```

HW2-1: Requirements (12/12)

- Requirements for rollback:
 - The snapshot should **still exist** after rollback
 - The subvolume **should be writable** after rollback
 - The rollbacked subvolume **state** should be **persist after umount/re mount**

HW2-1: Grading

- Shrink LV (3%)
- BTRFS-setup (9%)
 - BTRFS on LVM
 - subvolume structure
 - mounting layout
- LVM-Raid10 (5%)
- Snapper-Create (8%)
- Snapper-List (7%)
- Snapper-Delete (5%)
- Snapper-Rollback (11%)

HW2-1: Hint

- [mount\(8\)](#)
- [fstab\(5\)](#)
- [LVM - Archwiki](#)
- [BTRFS - Archwiki](#)

(2-2) (27%)

Manage SFTP file server

HW2-2: Overview

- Fix SSH settings
- Build a file server with SFTP
- Create a **administrator** with full access to the file server
- Create 2 **registered users** with limited access
 - administrator **is not counted** as registered users
- Create a readonly **anonymous** user
- Bury some treasure in your file server 🕵️

HW2-2: Requirements (1/5)

- SSH service should listen only on port 22, and disallow root login
- There should **NOT** be any banner
 - **DO NOT MODIFY OR DELETE** /etc/ssh/banner.txt
- There should **NOT** be **uname** output
- The greeting message should stay the same
- There should be **Last Login** message

HW2-2: Requirements (1/5)

```
root@pve:~# ssh sa
```

```
*****  
* This is the banner message, which should not be displayed.  
* DO NOT DELETE OR MODIFY THE FILE (/etc/ssh/banner.txt)  
* You will fail the testcase if you violate the rules.  
*****
```

These two should be deleted

```
Linux debian 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64
```

```
Ciallo~ (∠· 0< )☆
```

```
Welcome to SA HW2! Take your time, and have fun (probably with anxiety) doing this~
```

```
If you encounter any technical issue or have questions about the service,  
Please go to google groups. Good Luck!
```

```
judge@debian:~$ |
```

There should be Last Login: ..

HW2-2: Requirements (1/5)

```
root@pve:~# ssh judge@10.4.2.49
Ciallo~ (∠· 0< )~★
Welcome to SA HW2! Take your time, and have fun (probably with anxiety) doing this~

If you encounter any technical issue or have questions about the spec,
Please go to google groups. Good Luck!
Last login: Mon Sep 14 11:25:47 2026 from 10.4.2.48
judge@sa2026:~$ |
```

HW2-2: Requirements (2/5)

- Create following users:
 - `sysadm` (referred as **administrator**)
 - `sftp-u1`, `sftp-u2` (referred as **registered users**)
 - `anonymous`
- **Registered users** and **anonymous cannot login via SSH**
 - SFTP only
- Everyone should support login to sftp with ssh key
 - same public key with judge

HW2-2: Requirements (3/5)

- Host SFTP server on `$SFTP_ROOT`.
 - `$SFTP_ROOT = /mnt/hw2/sftp`
- Create 2 directories under `$SFTP_ROOT`, `/public` and `/private`
- **Registered users** and **anonymous** should **chroot** to `$SFTP_ROOT`
- **Administrator** should **not chroot**
- The **SFTP start directory** of **All user** should at `$SFTP_ROOT`

HW2-2: Requirements (4/5)

- \$SFTP_ROOT/public
 - **Registered user** can `get/put` file and `mkdir` in `public/**/*`
 - **Registered user** can only `rm/rmdir` **file/dir owned by them** in `public/**/*`
 - **anonymous** can only `get` file in `public/**/*` (readonly)
 - **Administrator** can `get/put/mkdir/rm/rmdir` **all content** in `public/**/*`
 - Every uploaded file should **remove** others' Read/Write/Execute permissions

NOTE1: `public/**/*` → all file under `public/`, includes its **subdirectories**

NOTE2: You may assume that registered user will **NOT** try to `rm/put` file under **subdirectories** of public **owned by other**

HW2-2: Requirements (5/5)

- `$SFTP_ROOT/private`
 - Create directory "`hidden/`" under `private/`
 - Put a file "`treasure`" under `hidden/`
 - **Registered user** and **anonymous**:
 - **cannot** perform `ls` in `private/**/*`
 - **cannot** write (`put/mkdir/rm/rmdir`) in `private/**/*`
 - can `cd` into `private/**/*`
 - can `get` "`treasure`" in `private/hidden/`
 - **Administrator** can `get/put/mkdir/rm/rmdir` **all content** in `private/**/*`

NOTE1: `private/**/*` ➡ all file under `private/`, includes its **subdirectories**

HW2-2: Quick reference

✓	✗	⚠
Allowed	Not allowed	only allowed for file owner itself

	sysadm		sftp-u1/2		anonymous	
	public/	hidden/	public/	hidden/	public/	hidden/
ls	✓	✓	✓	✗	✓	✗
mkdir	✓	✓	✓	✗	✗	✗
rmdir	✓	✓	⚠	✗	✗	✗
put	✓	✓	✓	✗	✗	✗
get	✓	✓	✓	✓	✓	✓
rm	✓	✓	⚠	✗	✗	✗

HW2-2: Grading

- SSH config fixing (2%)
- SFTP login only permission & chroot (3%)
- Sysadm access:
 - public (4%) private (3%)
- Sftp-u1/2 access:
 - public (4%) private (4%)
- anonymous access
 - public (3%) private (2%)
- Remove other's permission on uploaded file (2%)

HW2-2: Hint

- [sshd_config\(5\)](#)
- [sftp-server\(8\)](#)
- [chmod\(1\)](#)
- You can make following assumptions:
 - **registered user** will never **put/rm** file under subdirectories.
 - unless the subdirectories is owned by them

(2-3) (25%)
SFTP logging
and Monitoring daemon

HW2-3: Overview

- Enable **logging SFTP service** with both **journald** and **syslog**
- Program a **daemon** to **monitor and logging** suspicious action of SFTP user

HW2-3: Requirements (1/5)

- Enable **SFTP logging**, and user should be able to **access the log** via:
 - `journalctl -t internal-sftp -t sftp-server`
 - `cat /var/log/sftp.log`
- The log should be **pure SFTP log**
 - can't blend with log from other service (e.g. SSH, sudo...)

```
$ sudo journalctl -t sftp-server -t internal-sftp | tail -n 2
Oct 08 18:42:58 sa2025 sftp-server[1979]: session closed for local user sysadm from [10.0.2.2]
Oct 08 18:43:27 sa2025 internal-sftp[2029]: session opened for local user sftp-u1 from [10.0.2.2]
$ cat /var/log/sftp.log | tail -n 2
Oct 08 18:42:58 sa2025 sftp-server[1979]: session closed for local user sysadm from [10.0.2.2]
Oct 08 18:43:27 sa2025 internal-sftp[2029]: session opened for local user sftp-u1 from [10.0.2.2]
```

HW2-3: Requirements (2/5)

- Write a daemon that would watch on **every file SFTP user uploaded**.
 - Named as **sftp_watchd**.
 - The program can be written in any language (e.g. B, C, C++, C#, D, Python)
- **sftp_watchd** should reside in your system **\$PATH**
- Requirement of **sftp_watchd**:
 - Checking if any **uploaded file violate the rule**
 - If found, the program should **move** it into **\${SFTP_ROOT}/private/.violated/**
 - And **logging** it with specified format (detail in page 18)

HW2-3: Requirements (3/5)

- File violation rule:
 - a. Uploading **ELF** (Executable and Linkable format) file is prohibited
 - You may only check the first 4 bytes to identify a ELF file
 - b. Uploading **file with specific MD5 hash** is prohibited
 - Prohibited MD5 hash list:

```
209c6ec9c78249031b49b29aef2ee264
288d9c9c945b95bcc9a9632a594f8ebfc
84cbc60c4b110591d7c287da21067e70
d214f689364c2c19bd02aeb087a354e2
8806b1882ec4ee5f88f4e11641965285
```

HW2-3: Requirements (4/5)

- Logging message format:
 - a. ELF file

```
File {file} uploaded by user {user} is an ELF file. Moving to violated directory
```

- b. Prohibited MD5

```
File {file} uploaded by user {user} matches prohibited MD5 hash. Moving to violated directory
```

HW2-3: Requirements (5/5)

- Make **sftp_watchd** a **systemd service**
 - Writing a service unit file
- Enabling control **sftp_watchd** through **systemctl**
 - Should support start, stop, status, restart
- The log of **sftp_watchd** should be **accessed through journalctl**
 - `journalctl -u sftp_watchd`

HW2-3: Grading

- SFTP logging (6%)
- sftp_watchd functionality (12%)
- Managed sftp_watchd through systemd (7%)

HW2-3: Hint

- [Filter - Rsyslog documentation](#)
- [SFTP chroot - Archwiki](#)

Attention

- Deadline: 2026/12/21 (Mon.) 23:59
- Your work will be scored by Online Judge system
 - Only the **LAST submission** will be **scored**
 - Late submission will **NOT** be accepted
- We will fetch your script from `$(which sftp_watchd)` and `$(which SnApper)`
 - Make sure your script is readable by **judge** user from these path
 - Also, your script must not invoke any other self-written scripts, binaries or executables.

Attention

- **ALWAYS BACKUP** your system before submission
 - We may do malicious actions (e.g. `dd if=/dev/zero of=/dev/sda`)
- TAs reserve the right of final explanations.
 - Specs and the points of each subjudges are subject to change in any time.(with notification)
- Make sure everything works after reboot